

SheetEX v2 — Teacher Guide

Data wrangling for AI across Excel 365, Excel 2013, Google Sheets, CSV/TSV, SQL and relational databases.

10 lessons · 93 practice challenges · 152 certification-test questions · about 12 class periods (50 min) · certificates for every lesson

Teacher-only: this guide contains the answer key. It is NOT part of the student app.

How it runs

- **No student data is collected.** The Apps Script only serves the page. Progress lives in each student's own browser and in the codes they choose to turn in. Nothing is sent to you, to Google Drive or to anyone else.
- **Names are permanent.** On first visit a student types a first and last name and confirms it; after that it cannot be changed. A misspelled name means "Erase everything and start over" (all progress is lost), so check spelling on day one.
- **Practice, then test.** Practice challenges have hints. When a lesson's practice is done, the **certification test** unlocks: 10 random questions from the bank below (half hands-on), no hints, no cheat sheet, 80% to pass. An attempt counts when it starts (closing the page = a failed attempt), and a failed attempt has a 2-minute review break.
- **Certificates** show the test score, the attempt number, hints used in practice, and a verification ID. Students print/save as PDF, download an image, or paste the certificate code (`SXC2-...`) into Canvas.
- **Sealed codes.** Progress codes (`SXP2-...`) and certificate codes are signed with your deployment's class key (created automatically the first time anyone opens your web app). An edited code, a code with a changed name, or a code from any other copy of SheetEX is rejected. Run `newClassKey` once in the script editor to invalidate every old code (for example, each semester).
- **Checking work:** in the app click ? ► *For teachers:* paste all submitted codes at once, or type a name + lesson + verification ID from a printed certificate.
- **Honest limit:** the signing key has to live inside the page for the page to sign codes. This stops editing codes and saves; a student who reads the page's JavaScript in developer tools could, with real effort, rebuild the signing. On managed Chromebooks, Google Admin can turn developer tools off (Chrome policy *DeveloperToolsAvailability*). Treat certificates as strong evidence, like a screenshot of a proctored quiz, not cryptographic proof.

Suggested pacing

#	Lesson	Challenges	XP	Time
1	Spreadsheet Foundations Excel 365	10	225	50 min

2	Classic Excel & Nested Functions Excel 2013	8	190	50 min
3	Google Sheets Survival Guide Google Sheets	9	205	50 min
4	Data Files: CSV & TSV CSV / TSV	8	135	45 min
5	Data Wrangling I: Cleaning Data for AI Data Wrangling Lab	11	220	90 min
6	Data Wrangling II: Combining & Lookups Data Wrangling Lab	11	220	90 min
7	SQL for Data Analysts SQL Database	9	215	75 min
8	Cross-Platform Translator Compare	5	60	25 min
9	Databases 101: What is an RDBMS? RDBMS Explorer	10	175	60 min
10	Databases for Machine Learning ML Data Lab	12	240	75 min

Lessons 1–4 can be done in any order. Lessons 5–6 (Data Wrangling) are the core of the AI-data unit — the 🤖 AI check tab shows students how their cleaning changes what a model learns. Lesson 8 makes a good review day.

Differentiation: early finishers can switch the Data Wrangling Lab engine (Excel 365 ⇌ Google Sheets) and redo it in the other app, or explore the Interactive Cheat Sheet's SQL column.

Lesson 1: Spreadsheet Foundations

Workspace: **Excel 365** · 10 challenges · about 50 minutes

Spreadsheets are where most data work starts — including the data used to train and test AI. Modern Excel can answer questions about thousands of rows with one formula.

Students will be able to

- SUM, COUNTIF, SUMIF
- XLOOKUP
- FILTER, SORT, UNIQUE (dynamic arrays)
- IFS and LET
- Absolute vs relative references

Common misconceptions

- #SPILL! means something is in the way of a spilling result — clear the cells below.
- Fill formulas down with Ctrl+D; check the \$ signs before you do.
- A typed number is frozen; a formula updates when the data changes.

Vocabulary

Formula	Starts with = and calculates something.	<code>=SUM(F2:F25)</code>
Range	A rectangle of cells, written first:last.	<code>Products!D2:D25</code>
Relative vs absolute reference	A2 shifts when copied; \$A\$2 stays put.	<code>=E2*\$H\$1</code>
Function argument	The values inside the parentheses, separated by commas.	<code>SUMIF(range, criteria, sum_range)</code>
Dynamic array / spill	One formula returns many results that flow into the cells below.	<code>=FILTER(B2:B25, C2:C25="Drinks")</code>
Lookup	Find a row by a key and return another column from it.	<code>=XLOOKUP("SKU-404", A2:A25, B2:B25)</code>

Challenges & answer key

#	Challenge	Task	Answer
1	Your First Formula ★☆☆ · 10 XP	In Scratch!B2, find the total number of items in stock (the InStock column on the Products sheet). <i>Takeaway: SheetName!Range lets one sheet use data from another sheet.</i>	Scratch!B2 <code>=SUM(Products!F2:F25)</code>

2	What Is It All Worth? ★★★ · 15 XP	In Scratch!B3, calculate the total value of the inventory: every product's Price × InStock, all added up — in ONE formula. Takeaway: SUMPRODUCT works on every version. SUM(D2:D25*F2:F25) only works in Excel 365 (or with Ctrl+Shift+Enter in old Excel).	Scratch!B3 <pre>=SUMPRODUCT(Products!D2:D25,Products!F2:F25)</pre> Also accepted: <pre>=SUM(Products!D2:D25*Products!F2:F25)</pre>
3	Count the Categories ★★★ · 15 XP	In Scratch!B4, count how many products are in the School Supplies category. Takeaway: COUNTIF and SUMIF work the same in every spreadsheet app.	Scratch!B4 <pre>=COUNTIF(Products!C2:C25,"School Supplies")</pre>
4	Store S03 Sales ★★★ · 20 XP	In Scratch!B5, add up the Qty of every sale made by store S03 (Sales sheet). Takeaway: SUMIF = "add these numbers, but only on rows where something is true." In SQL it is SUM(...) WHERE ...	Scratch!B5 <pre>=SUMIF(Sales!C2:C61,"S03",Sales!E2:E61)</pre>
5	XLOOKUP Detective ★★★ · 20 XP	Type SKU-404 into Scratch!A7. In Scratch!B7, use XLOOKUP to show that SKU's product name. It should still work if someone types a different SKU into A7! Takeaway: XLOOKUP only exists in Excel 365/2021 and Google Sheets. Excel 2013 shows #NAME? — try it there next!	Scratch!A7 SKU-404 Scratch!B7 <pre>=XLOOKUP(A7,Products!A2:A25,Products!B2:B25)</pre>
6	Spill the Drinks ★★★ · 25 XP	In Scratch!D2, write ONE formula that lists the names of every product in the Drinks category. The answers should spill down. Takeaway: If anything is typed in the cells below D2, Excel shows #SPILL! because the answer has no room.	Scratch!D2 <pre>=FILTER(Products!B2:B25,Products!C2:C25="Drinks")</pre>
7	Sorted & Unique ★★★ · 25 XP	In Scratch!F2, list each product category once, sorted A→Z, in one formula. Takeaway: Nesting = putting a function inside another function. The inside one runs first.	Scratch!F2 <pre>=SORT(UNIQUE(Products!C2:C25))</pre>

8	Stock Status with IFS ★★★ · 30 XP	On the Products sheet, type Status in I1. In I2:I25 show "Reorder" if InStock is below ReorderAt, "Low" if InStock is below ReorderAt × 1.5, otherwise "OK". Use IFS. <i>Takeaway: IFS is new. In Excel 2013 you would need IF inside IF inside IF.</i>	Products!I1 Status Products!I2:I25 (type in the first cell, fill down) <code>=IFS(F2<G2,"Reorder",F2<G2*1.5,"Low",TRUE,"OK")</code>
9	LET It Be ★★★ · 30 XP	In Scratch!B9, calculate the profit margin of SKU-502: (price – cost) ÷ price. Use LET to name the price and the cost. <i>Takeaway: LET makes long formulas readable — and Excel only calculates each name once.</i>	Scratch!B9 <code>=LET(p,XLOOKUP("SKU-502",Products!A2:A25,Products!D2:D25),c,XLOOKUP("SKU-502",Products!A2:A25,Products!E2:E25),(p-c)/p)</code>
10	Build a Revenue Column ★★★ · 35 XP	On the Sales sheet, type Revenue in G1. Fill G2:G61 with Qty × Price. The price is on the Products sheet, so you will need a lookup. <i>Takeaway: \$A\$2 is an ABSOLUTE reference: it does not change when copied. A2 is RELATIVE: it shifts.</i>	Sales!G1 Revenue Sales!G2:G61 (type in the first cell, fill down) <code>=E2*XLOOKUP(D2,Products!\$A\$2:\$A\$25,Products!\$D\$2:\$D\$25)</code>

Certification test bank (16 questions; each test draws 10, half hands-on; 80% to pass)

#	Topic	Question	Answer
1	SUM hands-on	Write a formula for the total number of units in stock across all products (InStock is column F, rows 2–25).	<code>=SUM(Products!F2:F25)</code> Excel 365 · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
2	COUNTIF hands-on	Write a formula that counts how many products have a Price greater than \$10.	<code>=COUNTIF(Products!D2:D25,">10")</code> Excel 365 · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
3	SUMIF hands-on	Write a formula for the total Qty sold by store S02 (Sales rows 2–61).	<code>=SUMIF(Sales!C2:C61,"S02",Sales!E2:E61)</code> Excel 365 · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
4	AVERAGEIF hands-on	Write a formula for the average Price of the Snacks category.	<code>=AVERAGEIF(Products!C2:C25,"Snacks",Products!D2:D25)</code> Excel 365 · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.

5	XLOOKUP hands-on	Write a lookup formula that returns the Price of SKU-404. (It must keep working if rows are re-sorted.)	<code>=XLOOKUP("SKU-404",Products!A2:A25,Products!D2:D25)</code> Excel 365 · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
6	FILTER hands-on	Write ONE formula that spills the names of every Apparel product, in the order they appear in the table.	<code>=FILTER(Products!B2:B25,Products!C2:C25="Apparel")</code> Excel 365 · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
7	SORT + UNIQUE hands-on	Write ONE formula that spills an A→Z list of the categories, each listed once.	<code>=SORT(UNIQUE(Products!C2:C25))</code> Excel 365 · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
8	Relative references hands-on	In Products!J2, write a formula for that product's inventory value (Price × InStock). It will be filled down to row 25.	<code>=D2*F2</code> Excel 365 · Products!J2 filled to row 25. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
9	IF hands-on	In Products!J2, write a formula that shows "Reorder" when InStock is below ReorderAt, otherwise "OK". It will be filled down to row 25.	<code>=IF(F2<G2,"Reorder","OK")</code> Excel 365 · Products!J2 filled to row 25. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
10	#SPILL! concept	A FILTER formula shows #SPILL!. What is the most likely cause?	Cells below or beside it are not empty Other options: The formula has a typo in the function name · No rows matched the condition · The range is too small
11	Absolute references concept	Cell K2 holds =E2*\$H\$1. You fill it down to K3. What does K3 hold?	<code>=E3*\$H\$1</code> Other options: =E3*\$H\$2 · =E2*\$H\$1 · =F2*\$H\$1
12	Text functions concept	What does =LEN("Peachtree") return?	9
13	ROUND concept	What does =ROUND(12.4867, 2) return?	12.49
14	Dynamic arrays concept	Which of these functions returns a list of results that spills into the cells below?	UNIQUE Other options: COUNTIF · VLOOKUP · ROUND

15	Formulas vs typed values concept	Why is =SUM(F2:F25) better than typing the total (1,357) into the cell?	It updates when the data changes Other options: It makes the file smaller · Typed numbers cannot be formatted · Excel cannot add typed numbers
16	Fill down concept	You select J2:J25 (J2 has a formula) and press Ctrl+D. What happens?	J2 is copied into J3:J25, with references shifted row by row Other options: J2 is deleted · The column is sorted · J3:J25 get the exact same formula with no changes

Discussion

1. Why is a formula better than typing the answer, even if the answer is right today?
2. When would you use \$A\$2 instead of A2?

Exit ticket

Write a formula that counts how many products cost more than \$10.

Answer: `=COUNTIF(Products!D2:D25, ">10")`

Lesson 2: Classic Excel & Nested Functions

Workspace: **Excel 2013** · 8 challenges · about 50 minutes

Not everyone has the newest software. Building answers from classic functions teaches you how lookups and logic really work — and your files will open on any computer.

Students will be able to

- INDEX + MATCH
- VLOOKUP exact match
- Nested IF
- IFERROR
- Ctrl+Shift+Enter array formulas

Common misconceptions

- Forgetting FALSE in VLOOKUP gives wrong answers with no error.
- Without Ctrl+Shift+Enter, array formulas quietly use only one row.
- Count your parentheses: every IF(needs a).

Vocabulary

Nesting	Putting one function inside another. The inside one runs first.	<code>=IF(A2>10, "Big", IF(A2>5, "Medium", "Small"))</code>
INDEX / MATCH	MATCH finds the position; INDEX returns the value at that position.	<code>=INDEX(B2:B25, MATCH("SKU-404", A2:A25, 0))</code>
Exact vs approximate match	VLOOKUP needs FALSE for exact matches.	<code>=VLOOKUP(x, A2:D25, 3, FALSE)</code>
Array formula (CSE)	A formula that works on whole ranges, confirmed with Ctrl+Shift+Enter.	<code>{=MAX(IF(C2:C25="Snacks", D2:D25))}</code>
#NAME?	The program does not know a function name — often because it is too new.	<code>=XL00KUP(...)</code> in Excel 2013

Challenges & answer key

#	Challenge	Task	Answer
1	Back to 2013 ☆☆☆ · 10 XP	In Scratch!B2, try a modern formula: <code>=XLOOKUP("SKU-101", Products!A2:A25, Products!B2:B25)</code> . What does Excel 2013 do? Takeaway: #NAME? means "I don't know that word." Old Excel has never heard of	Scratch!B2 <code>=XL00KUP("SKU-101", Products!A2:A25, Products!B2:B25)</code> The correct result is #NAME? — the point is to see it.

XLOOKUP.

2	INDEX + MATCH ★★☆ · 25 XP	In Scratch!B3, show the product name for SKU-404 using INDEX and MATCH nested together (no VLOOKUP). Takeaway: INDEX/MATCH works in EVERY version of Excel and Google Sheets. It is the universal lookup.	Scratch!B3 <pre>=INDEX(Products!B2:B25,MATCH("SKU-404",Products!A2:A25,0))</pre>
3	VLOOKUP's Trap ★★☆ · 20 XP	In Scratch!B4, use VLOOKUP to find the price of the Hoodie by its name. Make sure you get the RIGHT price! Takeaway: Always finish VLOOKUP with FALSE (exact match). Forgetting it is the #1 VLOOKUP bug in the world.	Scratch!B4 <pre>=VLOOKUP("Hoodie",Products!B2:D25,3,FALSE)</pre> Without FALSE the answer is \$5.99 (wrong) — a great discussion point.
4	The Nested IF Ladder ★★☆ · 30 XP	On Products, fill I2:I25 with "Reorder" (InStock below ReorderAt), "Low" (below ReorderAt × 1.5) or "OK". IFS doesn't exist here — nest IF inside IF. Takeaway: Every IFS can be rewritten as nested IFs. Count your parentheses — each IF(needs a).	Products!I2:I25 (type in the first cell, fill down) <pre>=IF(F2<G2,"Reorder",IF(F2<G2*1.5,"Low","OK"))</pre>
5	IFERROR Safety Net ★★☆ · 20 XP	Type any SKU in Scratch!A6. In Scratch!B6, show its product name — or the words Not found if the SKU doesn't exist. Takeaway: IFERROR catches any error. IFNA (Excel 2013+) only catches #N/A, which is safer for lookups.	Scratch!A6 SKU-101 Scratch!B6 <pre>=IFERROR(VLOOKUP(A6,Products!A2:B25,2,FALSE),"Not found")</pre>
6	The Ctrl+Shift+Enter Club ★★★ · 40 XP	In Scratch!B7, find the highest Price in the Electronics category. MAXIFS does not exist in 2013! Use MAX(IF(...)) and confirm with Ctrl+Shift+Enter. Takeaway: Before dynamic arrays, formulas that work on whole ranges needed Ctrl+Shift+Enter. Without it, Excel quietly used only ONE row.	Scratch!B7 <pre>=MAX(IF(Products!C2:C25="Electronics",Products!D2:D25))</pre> Ctrl+Shift+Enter Confirm with Ctrl+Shift+Enter. Plain Enter gives a wrong answer.
7	Build a Price Tag ★★☆ · 15 XP	In Scratch!B8, build the text Granola Bar - \$1.50 from Products!B2 and Products!D2. Takeaway: Formatting (\$) is not part of the number. TEXT(...) bakes the format into text.	Scratch!B8 <pre>=Products!B2&" - "&TEXT(Products!D2,"\$0.00")</pre>
8	Wildcard SUMIFS ★★★ · 30 XP	In Scratch!B9, add up the Qty of sales from store S02 for products whose SKU starts with SKU-3 (school supplies).	Scratch!B9 <pre>=SUMIFS(Sales!E2:E61,Sales!C2:C61,"S02",Sales!D2:D61,"SKU-3*")</pre>

Takeaway: SUMIFS/COUNTIFS have existed since Excel 2007, so they are safe to use everywhere.

Certification test bank (15 questions; each test draws 10, half hands-on; 80% to pass)

#	Topic	Question	Answer
1	INDEX + MATCH hands-on	This workbook is opened in Excel 2013. Write a formula that returns the Price of SKU-404.	<code>=INDEX(Products!D2:D25,MATCH("SKU-404",Products!A2:A25,0))</code> Excel 2013 · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
2	VLOOKUP hands-on	In Excel 2013, write a formula that returns the Product name for SKU-203.	<code>=VLOOKUP("SKU-203",Products!A2:H25,2,FALSE)</code> Excel 2013 · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
3	Nested IF hands-on	Excel 2013 has no IFS. In Products!J2, show "Reorder" if InStock < ReorderAt, "Low" if InStock < 1.5 × ReorderAt, otherwise "OK". It will be filled down to row 25.	<code>=IF(F2<G2,"Reorder",IF(F2<G2*1.5,"Low","OK"))</code> Excel 2013 · Products!J2 filled to row 25. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
4	Array formulas (CSE) hands-on	Excel 2013 has no MAXIFS. Write a formula for the largest single-order Qty sold by store S03.	<code>=MAX(IF(Sales!C2:C61="S03",Sales!E2:E61))</code> <i>Ctrl+Shift+Enter</i> Excel 2013 · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
5	SUMPRODUCT hands-on	In ONE cell, write a formula for the total inventory value: the sum of Price × InStock over every product.	<code>=SUMPRODUCT(Products!D2:D25,Products!F2:F25)</code> Excel 2013 · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
6	COUNTIF hands-on	Write a formula that counts the orders with a Qty of 5 or more.	<code>=COUNTIF(Sales!E2:E61,">=5")</code> Excel 2013 · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
7	Joining text hands-on	In Stores!G2 (Excel 2013), join City and State like Cumming, GA. It will be filled down to row 6.	<code>=B2&"", "&C2"</code> Excel 2013 · Stores!G2 filled to row 6. Any formula with the same results on the

original and on a shuffled copy of the data is accepted.

8	#NAME? concept	What does Excel 2013 show for =XLOOKUP("SKU-101",A2:A25,B2:B25)?	#NAME? Other options: #N/A · #VALUE! · Granola Bar
9	Array formulas (CSE) concept	How do you enter an array formula like {=MAX(IF(...))} in Excel 2013?	Press Ctrl+Shift+Enter Other options: Type the curly braces yourself · Press Enter twice · Wrap it in ARRAYFORMULA()
10	VLOOKUP concept	In =VLOOKUP("SKU-203", A2:H25, 2, FALSE), what does FALSE mean?	Find an exact match only Other options: Return the 2nd match · Search from the bottom up · Ignore capital letters
11	VLOOKUP concept	Someone leaves off the last argument of VLOOKUP on an unsorted list. What happens?	It does an approximate match and can quietly return the wrong row Other options: It always returns #N/A · Excel refuses the formula · It works exactly the same as FALSE
12	MATCH concept	Products!A2:A25 starts SKU-101, SKU-102, SKU-103... What does =MATCH("SKU-103",Products!A2:A25,0) return?	3
13	Array formulas (CSE) concept	In Excel 2013, =MAX(IF(C2:C61="S03",E2:E61)) is entered with plain Enter. What happens?	It does not look at the whole range: it uses one row (or shows #VALUE!), so the answer is wrong Other options: It works the same as Ctrl+Shift+Enter · It shows #SPILL! · Excel converts it to MAXIFS
14	Version differences concept	Which function exists in Excel 2013?	IFERROR Other options: IFS · TEXTJOIN · XLOOKUP
15	IFERROR concept	What does =IFERROR(VLOOKUP("SKU-999",A2:D25,4,FALSE),"Not found") show when SKU-999 is not in the list?	Not found Other options: #N/A · 0 · FALSE

Discussion

1. Your formula works at school but shows #NAME? at home. What are two possible reasons?
2. Why did people love SUMPRODUCT before dynamic arrays existed?

Exit ticket

Rewrite =XLOOKUP("SKU-201", A2:A25, B2:B25) so it works in Excel 2013.

Answer: `=INDEX(B2:B25, MATCH("SKU-201", A2:A25, 0))`

Lesson 3: Google Sheets Survival Guide

Workspace: **Google Sheets** · 9 challenges · about 50 minutes

Google Sheets runs on every Chromebook and is built for sharing. It looks like Excel, but its functions do not always take the same arguments — the #1 reason "it worked at school but not at home".

Students will be able to

- FILTER and SORT the Sheets way
- ARRAYFORMULA
- QUERY
- SPLIT and REGEXMATCH
- Open-ended ranges

Common misconceptions

- SORT(..., -1) sorts ASCENDING in Sheets — no error, wrong answer.
- CONCAT only joins two values in Sheets.
- A bad formula is saved and shows #ERROR! instead of a pop-up.

Vocabulary

ARRAYFORMULA	Lets normal formulas work on whole ranges and fill a column with one formula.	<code>=ARRAYFORMULA(D2:D25-E2:E25)</code>
FILTER conditions	Each condition is its own argument (AND).	<code>=FILTER(B2:B25, C2:C25="Snacks", D2:D25>1)</code>
is_ascending	SORT direction is TRUE/FALSE in Sheets.	<code>=SORT(B2:D25, 3, FALSE)</code>
QUERY	SQL-like questions inside a spreadsheet; columns are letters.	<code>=QUERY(A1:H25, "select C, sum(F) group by C")</code>
Open-ended range	E2:E means "E2 to the bottom".	<code>=SUM(Sales!E2:E)</code>

Challenges & answer key

#	Challenge	Task	Answer
1	Excel Habits, Sheets Rules ★☆☆ · 20 XP	In Scratch!B2, type the Excel-style formula <code>=FILTER(Products!B2:B25, Products!C2:C25="Apparel", "None")</code> . Read the error, then fix it so it lists the Apparel products.	Scratch!B2 <code>=FILTER(Products!B2:B25, Products!C2:C25="Apparel")</code>

Takeaway: Same function name, different arguments. In Sheets every extra FILTER argument is another condition.

2	Two Conditions ★★☆ · 25 XP	In Scratch!D2, list Snacks that cost more than \$1.00. Give FILTER the two conditions as two separate arguments (the Sheets way). Takeaway: Excel needs one combined test instead: (cond1)*(cond2). The translator panel shows both!	Scratch!D2 <code>=FILTER(Products!B2:B25,Products!C2:C25="Snacks",Products!D2:D25>1)</code>
3	Which Way Is Down? ★★☆ · 20 XP	In Scratch!F2, sort Products!B2:D25 by price, highest first. (Hint: Excel's -1 will NOT work here — try it and see!) Takeaway: In Sheets, -1 counts as TRUE (any non-zero number is TRUE) so it sorted the WRONG way with no error. Silent bugs are the worst bugs.	Scratch!F2 <code>=SORT(Products!B2:D25,3,FALSE)</code>
4	One Formula, 24 Answers ★★☆ · 30 XP	On Products, put ONE formula in I2 that fills I2:I25 with the profit per unit (Price – Cost). Use ARRAYFORMULA. Takeaway: In Excel 365 this just works without ARRAYFORMULA. In Excel 2013 it needs Ctrl+Shift+Enter across the whole range.	Products!I2 <code>=ARRAYFORMULA(D2:D25-E2:E25)</code>
5	QUERY: SQL Inside a Spreadsheet ★★★ · 35 XP	In Scratch!J2, use QUERY to show each category and its total InStock. Takeaway: QUERY is Google-only. It is basically SQL — head over to SQL mode to see the real thing.	Scratch!J2 <code>=QUERY(Products!A1:H25,"select C, sum(F) group by C")</code>
6	SPLIT the Manager ★★☆ · 15 XP	In Scratch!B10, split the manager name in Stores!E2 into first name (B10) and last name (C10). Takeaway: SPLIT is Google-only. Excel uses Data ► Text to Columns (or TEXTSPLIT in newer Excel 365).	Scratch!B10 <code>=SPLIT(Stores!E2," ")</code>
7	Ranges With No Bottom ★★☆ · 20 XP	In Scratch!B12, add up all Qty on the Sales sheet using an open-ended range that starts at row 2 and has no last row. Takeaway: Open ranges grow automatically as you add rows. In Excel you would use E:E or a Table.	Scratch!B12 <code>=SUM(Sales!E2:E)</code>
8	The CONCAT Gotcha ★★☆ · 15 XP	In Scratch!B13, build Cumming, GA from Stores!B2 and Stores!C2. (Try =CONCAT(Stores!B2, " ", Stores!C2) first and read the error!)	Scratch!B13 <code>=CONCATENATE(Stores!B2," ",Stores!C2)</code>

Takeaway: The same function name can have different rules in different apps.

Also accepted: `=Stores!B2&" "&Stores!C2`

9 **Pattern Hunter**
★★★ · 25 XP

In Scratch!B14, count how many product names contain a digit (0-9) using REGEXMATCH.

Takeaway: Regular expressions are a mini-language for patterns. Google Sheets has them built in.

Scratch!B14
`=SUMPRODUCT(--
REGEXMATCH(Products!B2:B25, "\d")
)`

Certification test bank (15 questions; each test draws 10, half hands-on; 80% to pass)

#	Topic	Question	Answer
1	FILTER hands-on	In Google Sheets, write ONE formula that lists the names of all Drinks products.	<code>=FILTER(Products!B2:B25, Products!C2:C25="Drinks")</code> Google Sheets · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
2	SORT hands-on	In Google Sheets, sort the block Products!B2:D25 by Price (its 3rd column), highest price first.	<code>=SORT(Products!B2:D25, 3, FALSE)</code> Google Sheets · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
3	ARRAYFORMULA hands-on	In ONE cell, write a Google Sheets formula that fills a whole column with Price × InStock for every product (rows 2–25).	<code>=ARRAYFORMULA(Products!D2:D25*Products!F2:F25)</code> Google Sheets · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
4	QUERY hands-on	Use QUERY on Products!A1:H25 (1 header row) to list the Product names in the Snacks category, with the header Product on top.	<code>=QUERY(Products!A1:H25, "select B where C = 'Snacks'", 1)</code> Google Sheets · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
5	SPLIT hands-on	In Products!J2, split the Supplier name into one word per cell (spilling to the right). It will be filled down to row 25.	<code>=SPLIT(H2, " ")</code> Google Sheets · Products!J2 filled to row 25. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
6	REGEXMATCH hands-on	Write a Google Sheets formula that counts how many product names contain a digit (0–9).	<code>=SUMPRODUCT(-- REGEXMATCH(Products!B2:B25, "[0-9]"))</code> Google Sheets · Scratch!B2. Any formula with the same results on the original and

			on a shuffled copy of the data is accepted.
7	Joining text hands-on	In Stores!G2 (Google Sheets), join City and State like Cumming, GA. It will be filled down to row 6.	<code>=B2&" , "&C2</code> Google Sheets · Stores!G2 filled to row 6. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
8	Silent differences concept	In Google Sheets, <code>=SORT(A2:B25, 2, -1)</code> sorts column 2...	Lowest to highest — -1 counts as TRUE (ascending) Other options: Highest to lowest · It shows #ERROR! · It does not sort at all
9	Errors concept	You type a formula with a missing quote mark. What does Google Sheets do?	Accepts it and shows #ERROR! in the cell Other options: Shows a pop-up and refuses to enter it · Fixes the quote for you · Shows #NAME?
10	Open-ended ranges concept	What does E2:E mean in Google Sheets?	Column E from row 2 down to the last row Other options: Only cell E2 · Columns E through Z · It is an error
11	Joining text concept	How many values can Google Sheets' CONCAT join?	Exactly 2 Other options: Any number · Up to 30 · Only ranges
12	FILTER concept	In Google Sheets, <code>=FILTER(B2:B25, C2:C25="Snacks", D2:D25>1)</code> — what is the third argument?	A second condition (both must be true) Other options: What to show if nothing matches · The sort order · The number of results
13	SPLIT concept	Into how many cells does <code>=SPLIT("snacks drinks apparel", " ")</code> spill?	3 or three
14	ARRAYFORMULA concept	What is ARRAYFORMULA for?	Making one formula work on whole ranges so it fills a column Other options: Entering formulas with Ctrl+Shift+Enter · Sorting arrays · Importing data from another file
15	QUERY concept	In <code>=QUERY(A1:H25, "select B where C = 'Snacks'", 1)</code> , what are B and C?	Column letters of the range Other options: Named ranges · Sheet names · Cell values

Discussion

1. Which is worse: a formula that errors, or one that silently gives the wrong answer? Why?
2. Name one thing Google Sheets can do that Excel cannot, and one the other way around.

Exit ticket

In Google Sheets, sort Products!B2:D25 by price, lowest first.

Answer: `=SORT(Products!B2:D25, 3, TRUE)`

Lesson 4: Data Files: CSV & TSV

Workspace: **CSV / TSV** · 8 challenges · about 45 minutes

CSV is how data moves between programs — and how most AI training data is shared. It is plain text: if you understand the rules, you can read and fix any dataset.

Students will be able to

- Values vs formulas
- Quoting and escaping
- Delimiters
- How spreadsheets change data on import

Common misconceptions

- Saving as CSV throws away formulas, formatting and other sheets.
- Opening a CSV in a spreadsheet can drop leading zeros and mangle long IDs.
- An extra comma in one row shifts every value after it.

Vocabulary

Delimiter	The character between values: comma (CSV) or tab (TSV).	SKU-101,Granola Bar,Snacks
Field / record	A field is one value; a record is one line (row).	4 fields: a,b,c,d
Quoting	Values containing the delimiter, quotes or line breaks are wrapped in "double quotes".	"Gel Pens, 4-pack"
Escaping	A quote inside a quoted value is written twice.	"12"" Ruler"
Type inference	When a program opens a CSV it GUESSES types — and can change your data.	05401 → 5401

Challenges & answer key

#	Challenge	Task	Answer
1	Where Did My Formula Go? ☆☆☆ · 15 XP	You save an Excel workbook as CSV. Cell B2 had =SUM(D2:D25) showing \$189.63. What is stored in the CSV file? Takeaway: CSV keeps the RESULT, not the formula. Re-open it and your formulas are gone. Try "Export from a spreadsheet" to see it happen.	Answer: 189.63

2	Export a Sheet ★☆☆ · 15 XP	Click Export from spreadsheet, choose the Excel 365 → Products sheet and export it as CSV. Look at the "What was lost" report. Takeaway: Any formulas you added on that sheet became plain numbers in the CSV.	Export from spreadsheet ► Excel 365 ► Products ► CSV.
3	Fix the Broken Comma ★☆☆ · 25 XP	Open broken_products.csv. One row has too many fields because a product name contains a comma. Fix it so every row has 8 fields. Takeaway: Quotes protect commas inside values. This is why TSV (tabs) is popular — names rarely contain tabs.	In broken_products.csv change Gel Pens, 4-pack to "Gel Pens, 4-pack"
4	Commas to Tabs ★☆☆ · 20 XP	Create a new file named stores.tsv that holds the same data as stores.csv, but separated by TAB characters. Takeaway: Same data, different delimiter. Spreadsheets can open both.	New file stores.tsv, paste stores.csv, Find & replace , → \t (Replace all).
5	The Vanishing Zero ★☆☆ · 15 XP	Open suppliers.csv and switch the preview to "Open like a spreadsheet would". What happens to Lakeside Apparel's ZIP code 05401? Takeaway: ZIP codes, phone numbers and IDs are TEXT, not numbers. Import them as text or the zeros disappear forever.	Answer: It becomes 5401
6	Quotes Inside Quotes ★★★ · 25 XP	Add a new product to the end of products.csv: SKU SKU-307, product 12" Ruler (that's 12 inches!), category School Supplies, price 1.99, cost 0.6, InStock 30, ReorderAt 10, supplier Office Hub Supply. Takeaway: CSV escaping rule: wrap in quotes, double any inner quotes.	Add to products.csv: SKU-307,"12"" Ruler",School Supplies,1.99,0.6,30,10,Office Hub Supply
7	Count the Fields ★☆☆ · 10 XP	How many fields (values) are in this CSV line?SKU-302,"Gel Pens, 4-pack",School Supplies,4.5 Takeaway: SKU-302 Gel Pens, 4-pack School Supplies 4.5 → 4 fields.	Answer: 4
8	Pick the Right Format ★☆☆ · 10 XP	Which file type keeps your formulas, formatting and multiple sheets? Takeaway: Use .xlsx (or a Google Sheet) for working files. Use CSV/TSV to move data between programs.	Answer: .xlsx

Certification test bank (14 questions; each test draws 10, half hands-on; 80% to pass)

#	Topic	Question	Answer
1	Quoting commas hands-on	Write the CSV line for this row — SKU: SKU-302 · Product: Gel Pens, 4-pack · Price: 4.5	SKU-302,"Gel Pens, 4-pack",4.5
2	Escaping quotes hands-on	Write the CSV line for — SKU: SKU-307 · Product: 12" Ruler · Price: 1.99	SKU-307,"12"" Ruler",1.99
3	Empty fields hands-on	Write the CSV line for a row whose middle value is missing — SKU: SKU-999 · Product: (empty) · Price: 2.5	SKU-999,,2.5
4	TSV hands-on	Write the TSV (tab-separated) line for — Product: Gel Pens, 4-pack · Price: 4.5. (Press Tab in the answer box to type a tab.)	Gel Pens, 4-pack↵4.5
5	Quoting commas hands-on	Write a 2-line CSV file: a header row Name, Note and one data row — Name: Avery · Note: Said "hi", then left	Name,Note Avery,"Said ""hi"", then left"
6	Leading zeros hands-on	Write the CSV line for — City: Burlington · State: VT · Zip: 05401 (keep the zero!)	Burlington,VT,05401
7	Values vs formulas concept	A sheet with =SUM(D2:D25) in D26 is saved as CSV. What is stored in that spot?	Only the result, like 233.12 Other options: The formula =SUM(D2:D25) · Both the formula and the result · Nothing — CSV skips formula cells
8	Leading zeros concept	A CSV holds the Zip 05401. You open it by double-clicking in Excel. What do you see?	5401 Other options: 05401 · #VALUE! · 5,401.00
9	Long numbers concept	A CSV has a 16-digit ID 4111222233334444. After opening and saving in Excel it becomes...	4111222233334440 (Excel keeps only 15 digits) Other options: Exactly the same · #NUM! · The text "4111222233334444"
10	Delimiters concept	What separates the values in a TSV file?	A tab character Other options: A comma · A semicolon · A space
11	Quoting commas concept	How many fields are in this CSV line? a,"b,c",d	3 or three

12	Values vs formulas concept	Which of these does a CSV file keep?	The values in one sheet Other options: Bold text and colors · Several sheets · Formulas
13	Escaping quotes concept	Inside a quoted CSV value, how do you write ONE quote mark character? Type exactly what goes in the file.	<code>""</code>
14	Import surprises concept	A CSV contains the code 1-2 (aisle 1, shelf 2). Opening it in Excel shows...	A date like 2-Jan Other options: 1-2 · -1 · #VALUE!

Discussion

1. You email a coworker a CSV of your gradebook. What did you lose?
2. Why would anyone choose TSV over CSV?

Exit ticket

Write the CSV line for: Smith, Jo | 05401 | said "hi"

Answer: `"Smith, Jo",05401,"said ""hi"""`

Lesson 5: Data Wrangling I: Cleaning Data for AI

Workspace: **Data Wrangling Lab** · 11 challenges · about 90 minutes

Data scientists spend most of their time cleaning data. An AI model learns whatever is in its training data — including typos, duplicates and fake zeros. Clean data is the difference between a useful model and a confident wrong one.

Students will be able to

- Leading zeros and text vs numbers
- Finding blank and fake-null values
- Converting text to numbers
- Fixing mixed date formats
- TRIM, PROPER, UPPER, nested text formulas

Common misconceptions

- Replacing missing values with 0 changes averages — missing is not zero.
- 03.09.2026 is September 3 in Europe and March 9 in the US.
- Clean into NEW columns so you can always compare with the original.

Vocabulary

Data type	Number, text, date, TRUE/FALSE. Digits can be TEXT ("0104").	<code>=ISTEXT(C2)</code>
Null / missing value	No value. Can hide as blanks or as words like N/A, -, null.	<code>=COUNTBLANK(E2:E31)</code>
Type conversion	Changing a value's type.	<code>=VALUE("12") =TEXT(104, "0000")</code>
Normalization	Making the same thing look the same: case, spaces, formats.	<code>=UPPER(TRIM(B2))</code>
Date serial	Real dates are numbers (days since 1900), so they sort and subtract.	<code>=DATEVALUE("Sep 4, 2026")</code>
Garbage in, garbage out	A model can only be as good as its data.	Open the  AI check tab

Challenges & answer key

#	Challenge	Task	Answer
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1	Where Did the Zero Go? ★★★ · 10 XP	RawOrders!C2 shows 104, but the real item code is 0104. Why? Takeaway: IDs, ZIP codes and phone numbers are TEXT, even though they are made of digits. An AI model will treat 104 and "0104" as two different items.	Answer: It was stored as a NUMBER, and numbers never keep leading zeros
2	Rebuild the Item Codes ★★★ · 20 XP	Type CleanCode in RawOrders!I1. In I2:I31, turn every ItemCode (column C) into a 4-character text code like 0104 — whether it arrived as 104, "0104" or "0402". Takeaway: TEXT(x, "0000") is the spreadsheet version of zero-padding. SQL: printf('%04d', x). Python: str(x).zfill(4).	RawOrders!I1 CleanCode RawOrders!I2:I31 (type in the first cell, fill down) =TEXT(C2, "0000")
3	Count the Holes ★★★ · 10 XP	In Report!B2, count how many Qty cells (RawOrders!E2:E31) are completely empty. Takeaway: Empty is only ONE kind of missing data. Look at the next challenge...	Report!B2 =COUNTBLANK(RawOrders!E2:E31)
4	Fake Nulls ★★★ · 20 XP	Some systems write missing values as words. In Report!B3, count Qty cells that say N/A, - or null. Takeaway: COUNTBLANK said 2 — the real number of missing quantities is 5. "null", "N/A" and "-" are TEXT that MEANS missing. Before training a model, they must all become one true blank/NA.	Report!B3 =COUNTIF(RawOrders!E2:E31, "N/A")+COUNTIF(RawOrders!E2:E31, "-")+COUNTIF(RawOrders!E2:E31, "null")
5	One Kind of Missing ★★★ · 25 XP	Type CleanQty in J1. In J2:J31, turn Qty into a real number — "12" (text) becomes 12 — and turn every kind of missing value into a blank "". Takeaway: Text that looks like a number does not add up: SUM ignores "12". Converting types is step one of every data-cleaning job.	RawOrders!J1 CleanQty RawOrders!J2:J31 (type in the first cell, fill down) =IFERROR(VALUE(E2), "")
6	Same Store, Different Spelling ★★★ · 15 XP	Type CleanStore in K1. In K2:K31, clean the Store codes so "s04" and " S02 " become S04 and S02. Takeaway: To a computer "S04", "s04" and " S04" are three different stores. A model trained on this would think you have 13 stores!	RawOrders!K1 CleanStore RawOrders!K2:K31 (type in the first cell, fill down) =UPPER(TRIM(B2))
7	Prices That Are Really Text ★★★ · 15 XP	Type CleanPrice in L1. In L2:L31, turn UnitPrice into real numbers (" \$3.25 " → 3.25, "14.99 " → 14.99).	RawOrders!L1 CleanPrice

		Takeaway: If a column has even ONE text price, averages and sums silently skip it. Always check types.	RawOrders!L2:L31 (type in the first cell, fill down) <code>=VALUE(F2)</code>
8	Four Kinds of Dates ★★★ · 30 XP	Type CleanDate in M1. In M2:M31, make every OrderDate a real date. Some are already dates, others are text like "2026-09-12", "9/23/2026" or "Sep 4, 2026". (Row 19 is special — skip it for now.) Takeaway: Real dates are numbers (days since 1900), so you can sort them, subtract them and group them by month. Text dates cannot do any of that.	RawOrders!M1 CleanDate RawOrders!M2:M31 (type in the first cell, fill down) <code>=IF(ISNUMBER(D2),D2,DATEVALUE(D2))</code> Row 19 (European date) shows #VALUE! here — that is the next challenge.
9	The European Date ★★★ · 30 XP	RawOrders!D19 says 03.09.2026 — in Europe that means 3 September 2026. DATEVALUE cannot read it. In Report!B5, build the real date from its pieces. Takeaway: Is 03/09 March 9th or September 3rd? It depends on the country. Merging data from different places is one of the most common ways datasets go wrong.	Report!B5 <code>=DATE(RIGHT(RawOrders!D19,4),MID(RawOrders!D19,4,2),LEFT(RawOrders!D19,2))</code>
10	Fix the Names ★★★ · 35 XP	Type CleanName in N1. In N2:N31, turn every Customer into First Last with proper capitals: "jordan smith", " Avery Johnson ", "TAYLOR KIM" and even "Davis, Morgan" → "Morgan Davis". Takeaway: This is a nested formula: IF, ISNUMBER, FIND, PROPER, TRIM, MID and LEFT working together. Building it one piece at a time is the skill.	RawOrders!N1 CleanName RawOrders!N2:N31 (type in the first cell, fill down) <code>=IF(ISNUMBER(FIND(",",G2)),PROPER(TRIM(MID(G2,FIND(",",G2)+1,99))))&"</code> <code>"&LEFT(G2,FIND(",",G2)-1),PROPER(TRIM(G2))</code>
11	How Dirty Was It? ★★★ · 10 XP	After CleanQty is done, count in Report!B4 how many orders have no usable quantity. Takeaway: Report your missing data. A model trained without telling anyone that 1 in 6 rows was dropped can be badly biased.	Report!B4 <code>=COUNTBLANK(RawOrders!J2:J31)</code> Needs earlier step(s): wr-qty

Certification test bank (15 questions; each test draws 10, half hands-on; 80% to pass)

#	Topic	Question	Answer
1	Leading zeros hands-on	ItemCode (column C) is a mess of numbers like 104 and text like " 0402". In RawOrders!S2, turn it into a clean 4-	<code>=TEXT(C2,"0000")</code> Excel 365 · RawOrders!S2 filled to row 31. Any formula with the same results on

		character code like 0104. It will be filled down to row 31.	the original and on a shuffled copy of the data is accepted.
2	Text to numbers hands-on	In RawOrders!S2, convert Qty (column E) to a real number, and show a truly blank result ("") when there is no usable number (blank, N/A, -, null). Filled down to row 31.	<code>=IFERROR(VALUE(E2),"")</code> Excel 365 · RawOrders!S2 filled to row 31. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
3	TRIM / UPPER hands-on	Store codes (column B) look like " S02 ", s04, S01. In RawOrders!S2, clean them to the form S02. Filled down to row 31.	<code>=UPPER(TRIM(B2))</code> Excel 365 · RawOrders!S2 filled to row 31. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
4	Text to numbers hands-on	UnitPrice (column F) holds things like "\$3.25" and "14.99 ". In RawOrders!S2, turn it into a real number. Filled down to row 31.	<code>=VALUE(F2)</code> Excel 365 · RawOrders!S2 filled to row 31. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
5	TRIM / PROPER hands-on	In RawOrders!S2, clean Customer (column G): remove extra spaces and capitalize each word (" avery johnson " → Avery Johnson). Filled down to row 31.	<code>=PROPER(TRIM(G2))</code> Excel 365 · RawOrders!S2 filled to row 31. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
6	Missing values hands-on	Write a formula that counts the Qty cells (E2:E31) that are completely empty.	<code>=COUNTBLANK(RawOrders!E2:E31)</code> Excel 365 · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
7	Missing values hands-on	Write a formula that counts the Qty cells (E2:E31) holding a fake-null word: N/A, - or null.	<code>=COUNTIF(RawOrders!E2:E31,"N/A")+COUNTIF(RawOrders!E2:E31,"-")+COUNTIF(RawOrders!E2:E31,"null")</code> Excel 365 · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
8	Dates hands-on	RawOrders!D19 holds the European date text 03.09.2026 (day.month.year = 3 September 2026). Write a formula that builds a real date from that cell's text.	<code>=DATE(RIGHT(RawOrders!D19,4),MID(RawOrders!D19,4,2),LEFT(RawOrders!D19,2))</code> Excel 365 · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
9	Leading zeros concept	Why does it matter that item code 0104 became the number 104?	Lookups against the text code "0104" will not match Other options: Numbers take more memory · 104 is a different product · It does not matter

10	Missing values concept	Which of these is a fake null that should be cleaned into a real blank?	"N/A" Other options: 0.5 · "S02" · "0104"
11	TRIM / UPPER concept	What does =UPPER(TRIM(" s02 ")) return?	S02
12	Text to numbers concept	E2 holds the TEXT "12" and E3 holds the number 8. What does =SUM(E2:E3) return?	8 Other options: 20 · #VALUE! · 128
13	Missing values concept	Why not just replace every missing quantity with 0 before training an AI model?	The model learns that orders are smaller than they really are Other options: Zero is not a number · AI models cannot read 0 · It makes the file too big
14	Leading zeros concept	What does =TEXT(7, "000") return?	007
15	Text to numbers concept	What does =ISNUMBER("12") return?	FALSE Other options: TRUE · 12 · #VALUE!

Discussion

1. An AI model was trained on data where missing quantities were replaced with 0. What will it get wrong?
2. Is 03/04/2026 March 4 or April 3? How would you find out for a real dataset?
3. Why clean into a NEW column instead of overwriting the original?

Exit ticket

A column of ZIP codes shows 2110 and 5401. Write the formula that fixes them.

Answer: `=TEXT(A2, "00000")`

Lesson 6: Data Wrangling II: Combining & Lookups

Workspace: **Data Wrangling Lab** · 11 challenges · about 90 minutes

Real datasets live in separate tables. Joining them — matching orders to products, stores to regions — only works when the keys match exactly. This is the same idea as a JOIN in SQL or merge() in Python.

Students will be able to

- VLOOKUP and HLOOKUP
- Why lookups fail (type mismatch)
- Concatenation and delimiters
- TEXTJOIN
- Filtering around missing values

Common misconceptions

- Clean the key column BEFORE looking up.
- Lock lookup ranges with \$ before filling down.
- Blank text "" counts as bigger than any number in comparisons.

Vocabulary

Key	The column used to match rows between tables.	Item code 0104
VLOOKUP / HLOOKUP	Look up down the first column / across the first row.	=HLOOKUP("S03", Targets!A1:F3, 3, FALSE)
Type mismatch	Number 104 never equals text "0104" — the lookup returns #N/A.	=VLOOKUP(C2, ...) → #N/A
Concatenation	Joining text together.	=B2 & ", " & A2
Delimited list	Several values in one cell separated by a character.	snacks drinks apparel
Filtering	Keeping only rows that pass a test — watch out for blanks.	=FILTER(A2:A31, ISNUMBER(J2:J31)*(J2:J31>10))

Challenges & answer key

#	Challenge	Task	Answer
1	The #N/A Mystery ★☆☆ · 15 XP	=VLOOKUP(RawOrders!C2, ItemCodes!A2:D25, 2, FALSE) gives #N/A, even though code 0104 IS in the ItemCodes table. Why?	Answer: C2 is the NUMBER 104 and the table has the TEXT "0104" — they never match

Takeaway: A lookup only matches the same TYPE: number 104 ≠ text "0104". This is the #1 reason lookups fail on real data.

2	Look Up the Products ★★★ · 20 XP	Type Product in O1. In O2:O31, use VLOOKUP with your CleanCode (column I) to show each product name from ItemCodes. Takeaway: Clean first, then join. In SQL this is a JOIN; in Python pandas it is merge().	RawOrders!O1 Product RawOrders!O2:O31 (type in the first cell, fill down) =VLOOKUP(I2,ItemCodes!\$A\$2:\$D\$25,2,FALSE) Needs earlier step(s): wr-code
3	Line Totals ★★★ · 25 XP	Type LineTotal in P1. In P2:P31, calculate CleanQty × the ItemCodes price. If CleanQty is blank, leave LineTotal blank. Takeaway: Missing values spread: blank × price should stay blank, not turn into 0 — a fake 0 would teach a model that people order nothing.	RawOrders!P1 LineTotal RawOrders!P2:P31 (type in the first cell, fill down) =IF(J2="", "", J2*XL00KUP(I2,ItemCodes!\$A\$2:\$A\$25,ItemCodes!\$D\$2:\$D\$25)) Needs earlier step(s): wr-code, wr-qty
4	HLOOKUP Goes Sideways ★★★ · 15 XP	The Targets sheet is laid out sideways (stores across the top). In Report!B7, use HLOOKUP to find the Sept Target for store S03. Takeaway: VLOOKUP = vertical tables, HLOOKUP = horizontal tables. XLOOKUP and INDEX/MATCH handle both.	Report!B7 =HLOOKUP("S03",Targets!A1:F3,3,FALSE)
5	Region for Every Order ★★★ · 20 XP	Type Region in Q1. In Q2:Q31, use HLOOKUP with your CleanStore (column K) to show each order's region. Takeaway: Every lookup in this lab worked only because you cleaned the key column first.	RawOrders!Q1 Region RawOrders!Q2:Q31 (type in the first cell, fill down) =HLOOKUP(K2,Targets!\$B\$1:\$F\$2,2,FALSE) Needs earlier step(s): wr-store
6	Now the Math Works ★★★ · 10 XP	In Report!B8, add up all the cleaned quantities (CleanQty). Takeaway: SUM(E2:E31) gives a smaller, WRONG total because it skips text numbers. Clean data changes the answer.	Report!B8 =SUM(RawOrders!J2:J31) Needs earlier step(s): wr-qty
7	Last, First ★★★ · 15 XP	On Contacts, fill G2:G9 with names written as Last, First (e.g. "Whitfield, Dana"). Takeaway: A delimiter is the character that separates pieces of text — here it is ", ".	Contacts!G2:G9 (type in the first cell, fill down) =B2&"", "&A2
8	Keep the Zero (Again) ★★★ · 25 XP	In Contacts!H2:H9, build a mailing line like Burlington, VT 05401. Watch out:	Contacts!H2:H9 (type in the first cell, fill down)

		the Zip column lost its leading zeros! Takeaway: ZIP codes in New England start with 0. Treat ZIPs as text from the start and this problem never happens.	<code>=C2&" , "&D2&"</code> <code>"&TEXT(E2,"00000")</code>
9	Count the Delimited Tags ★★★ · 30 XP	Column F holds tags separated by a pipe: snacks drinks. In Contacts!I2:I9, count how many tags each contact has (blank = 0). Takeaway: Sheets has SPLIT(F2," ") and Excel 365 has TEXTSPLIT to break the tags into cells. Counting delimiters is the trick that works everywhere.	Contacts!I2:I9 (type in the first cell, fill down) <code>=IF(F2="",0,LEN(F2)-LEN(SUBSTITUTE(F2," ",""))+1)</code>
10	Write a CSV Line ★★★ · 15 XP	In Report!B10, join Contacts!A2:E2 into one CSV line with commas between the values. Takeaway: This is exactly what "Save as CSV" does for every row. Now you know why a comma INSIDE a value breaks the file.	Report!B10 <code>=TEXTJOIN(", ",FALSE,Contacts!A2:E2)</code>
11	Filter the Feed ★★★ · 30 XP	In Report!D2, use FILTER to list the OrderIDs whose CleanQty is more than 10. (Careful — try J2:J31>10 alone first and look at what sneaks in!) Takeaway: Missing values break filters in sneaky ways. Always ask "what happens to the blanks?"	Report!D2 <code>=FILTER(RawOrders!A2:A31,ISNUMBER(RawOrders!J2:J31)*(RawOrders!J2:J31>10))</code> Needs earlier step(s): wr-qty

Certification test bank (15 questions; each test draws 10, half hands-on; 80% to pass)

#	Topic	Question	Answer
1	VLOOKUP hands-on	In RawOrders!S2, look up the Product name for the row's ItemCode (column C) in ItemCodes!A2:D25 (codes there are text like 0104). Every row must find its product. Filled down to row 31.	<code>=VLOOKUP(TEXT(C2,"0000"),ItemCodes!\$A\$2:\$D\$25,2,FALSE)</code> Excel 365 · RawOrders!S2 filled to row 31. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
2	HLOOKUP hands-on	The Targets sheet runs sideways (stores across row 1). Write a formula that returns the Sept Target for store S03.	<code>=HLOOKUP("S03",Targets!A1:F3,3,FALSE)</code> Excel 365 · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
3	HLOOKUP hands-on	In RawOrders!S2, return the Region (Targets row 2) for the row's store — note the store codes in column B are messy. Filled down to row 31.	<code>=HLOOKUP(UPPER(TRIM(B2)),Targets!\$B\$1:\$F\$2,2,FALSE)</code> Excel 365 · RawOrders!S2 filled to row 31. Any formula with the same results on

			the original and on a shuffled copy of the data is accepted.
4	Concatenation hands-on	In Contacts!G2, build Last, First (like Whitfield, Dana). Filled down to row 9.	<code>=B2&" , "&A2</code> Excel 365 · Contacts!G2 filled to row 9. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
5	Concatenation hands-on	In Contacts!G2, build an address line like Cumming, GA 30040 — Zip codes must always show 5 digits. Filled down to row 9.	<code>=C2&" , "&D2&"</code> <code>"&TEXT(E2,"00000")</code> Excel 365 · Contacts!G2 filled to row 9. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
6	TEXTJOIN hands-on	Write ONE formula that joins all first names in Contacts!A2:A9 into one cell, separated by a comma and a space.	<code>=TEXTJOIN(" , ",TRUE,Contacts!A2:A9)</code> Excel 365 · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
7	FILTER hands-on	Write ONE formula that spills the OrderIDs whose Qty (column E) is already a real number.	<code>=FILTER(RawOrders!A2:A31,ISNUMBER(RawOrders!E2:E31))</code> Excel 365 · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
8	Delimiters hands-on	Tags (column F) are separated by , like snacks drinks. In Contacts!G2, count the number of tags (0 when empty). Filled down to row 9.	<code>=IF(F2="",0,LEN(F2)-LEN(SUBSTITUTE(F2," ",""))+1)</code> Excel 365 · Contacts!G2 filled to row 9. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
9	Why lookups fail concept	=VLOOKUP(C2, ItemCodes!A2:D25, 2, FALSE) shows #N/A when C2 is the number 104 and the table has "0104". Why?	The number 104 and the text "0104" are different values Other options: VLOOKUP cannot read numbers · The column number is wrong · The table must be sorted
10	HLOOKUP concept	Where does HLOOKUP search for the key?	Across the first row of the table Other options: Down the first column · In every cell · In the last row
11	TEXTJOIN concept	What does =TEXTJOIN("-", TRUE, "a", "", "b") return?	a-b

12	VLOOKUP concept	In =VLOOKUP(key, ItemCodes!A2:D25, 4, FALSE), which column is returned?	Column D (the 4th column of the table) Other options: Column 4 of the whole sheet only if the table starts in A · The 4th matching row · Column E
13	Delimiters concept	What is a delimiter?	A character that separates values, like , or or a tab Other options: A function that removes spaces · A type of number format · The header row
14	Absolute references concept	Why write ItemCodes!\$A\$2:\$D\$25 (with \$) in a VLOOKUP you will fill down?	So the lookup table does not shift as the formula moves down Other options: To format the result as money · Because VLOOKUP requires \$ · To make it faster
15	Concatenation concept	What does ="S" & TEXT(3, "00") return?	S03

Discussion

1. VLOOKUP says #N/A but you can SEE the value in the table. List three reasons this happens.
2. Why do joins (lookups) come AFTER cleaning, never before?

Exit ticket

Count the items in "red|green|blue" with one formula that works in every app.

Answer: `=LEN(A2) - LEN(SUBSTITUTE(A2, "|", "")) + 1`

Lesson 7: SQL for Data Analysts

Workspace: **SQL Database** · 9 challenges · about 75 minutes

Companies keep their data in databases, and SQL is how you ask them questions. Every spreadsheet skill has a SQL twin — and SQL scales to millions of rows.

Students will be able to

- SELECT, WHERE, ORDER BY
- GROUP BY and HAVING
- JOIN (the SQL VLOOKUP)
- Integer division
- UPDATE safely

Common misconceptions

- Text uses 'single quotes'.
- Whole-number division drops decimals: $7 / 2 = 3$.
- UPDATE or DELETE without WHERE changes every row.

Vocabulary

SELECT ... FROM	Choose columns from a table.	<code>SELECT product, price FROM products;</code>
WHERE	Keep only matching rows (like FILTER).	<code>WHERE category = 'Snacks'</code>
GROUP BY	One result per group (like SUMIF for every value at once).	<code>SELECT store_id, SUM(qty) FROM sales GROUP BY store_id;</code>
JOIN	Combine tables on a key (like VLOOKUP for every row).	<code>FROM sales s JOIN products p ON s.sku = p.sku</code>
NULL	SQL's missing value. Test with IS NULL, never = NULL.	<code>WHERE qty IS NULL</code>

Challenges & answer key

#	Challenge	Task	Answer
1	SELECT Everything ☆☆☆ · 10 XP	Show every column and every row of the products table. Takeaway: A spreadsheet shows all data all the time. A database only shows what you ask for.	<code>SELECT * FROM products;</code>

2	WHERE = FILTER ★★☆ · 15 XP	Show the product and price of every product in the Electronics category. Takeaway: WHERE works like FILTER. Remember: 'single quotes' for text, and it is case-sensitive.	<pre>SELECT product, price FROM products WHERE category = 'Electronics';</pre>
3	Top 3 Most Expensive ★★☆ · 20 XP	Show the product and price of the 3 most expensive products, highest first. Takeaway: ORDER BY = SORT. LIMIT = "only the first N".	<pre>SELECT product, price FROM products ORDER BY price DESC LIMIT 3;</pre>
4	GROUP BY = SUMIF for Everyone ★★☆ · 25 XP	Show each store_id with the total qty it sold (from the sales table). Takeaway: One GROUP BY query does what five SUMIF formulas would.	<pre>SELECT store_id, SUM(qty) FROM sales GROUP BY store_id;</pre>
5	JOIN = VLOOKUP ★★☆ · 30 XP	For every sale, show the order_id and the product name. The name lives in the products table, so you need to JOIN. Takeaway: A JOIN does a lookup for EVERY row at once — no filling down.	<pre>SELECT s.order_id, p.product FROM sales s JOIN products p ON s.sku = p.sku;</pre>
6	HAVING: Filter the Groups ★★★ · 30 XP	Show the categories that have more than 5 products, with their count. Takeaway: WHERE → rows. HAVING → groups.	<pre>SELECT category, COUNT(*) FROM products GROUP BY category HAVING COUNT(*) > 5;</pre>
7	Revenue by City ★★★ · 40 XP	Show each store's city and its total revenue (qty × price), highest first. You need sales, stores AND products. Takeaway: This would take lookups, a helper column and a pivot table in a spreadsheet. In SQL it is one query.	<pre>SELECT st.city, SUM(s.qty * p.price) AS revenue FROM sales s JOIN stores st ON s.store_id = st.store_id JOIN products p ON s.sku = p.sku GROUP BY st.city ORDER BY revenue DESC;</pre>
8	The Integer Division Trap ★★☆ · 20 XP	Find the average qty per sale with its decimals (it is NOT a whole number). Try SELECT SUM(qty) / COUNT(*) FROM sales; first and see what goes wrong. Takeaway: In SQL, 7 / 2 = 3. In a spreadsheet, 7 / 2 = 3.5. Same math, different rules!	<pre>SELECT AVG(qty) FROM sales;</pre> SUM(qty)/COUNT(*) gives 4 (integer division) — the trap.

9 **Restock Day**
★★★ · 25 XP

The truck arrived! For every product where `in_stock` is below `reorder_at`, set `in_stock` to `reorder_at × 2`. Leave the others alone. (If you mess up, use Reset database.)

Takeaway: Forget the `WHERE` and `EVERY` product changes. Real databases have no `Ctrl+Z`!

```
UPDATE products SET in_stock =  
reorder_at * 2 WHERE in_stock <  
reorder_at;
```

Certification test bank (15 questions; each test draws 10, half hands-on; 80% to pass)

#	Topic	Question	Answer
1	WHERE hands-on	List the product and price of every product in the Electronics category.	<pre>SELECT product, price FROM products WHERE category = 'Electronics';</pre> Any SELECT returning the same rows (column order does not matter) is accepted.
2	ORDER BY / LIMIT hands-on	List the product and price of the 3 most expensive products, most expensive first.	<pre>SELECT product, price FROM products ORDER BY price DESC LIMIT 3;</pre> Any SELECT returning the same rows in the same order (column order does not matter) is accepted.
3	GROUP BY hands-on	For each <code>store_id</code> , show the total qty sold (two columns: <code>store_id</code> , total).	<pre>SELECT store_id, SUM(qty) FROM sales GROUP BY store_id;</pre> Any SELECT returning the same rows (column order does not matter) is accepted.
4	JOIN hands-on	For every sale at store <code>S05</code> , show the <code>order_id</code> and the product name (two columns).	<pre>SELECT s.order_id, p.product FROM sales s JOIN products p ON s.sku = p.sku WHERE s.store_id = 'S05';</pre> Any SELECT returning the same rows (column order does not matter) is accepted.
5	HAVING hands-on	Show each category with more than 4 products and its product count (two columns).	<pre>SELECT category, COUNT(*) FROM products GROUP BY category HAVING COUNT(*) > 4;</pre> Any SELECT returning the same rows (column order does not matter) is accepted.
6	Aggregates hands-on	Show the average qty per sale (one value).	<pre>SELECT AVG(qty) FROM sales;</pre> Any SELECT returning the same rows (column order does not matter) is

			accepted.
7	JOIN hands-on	Show each city and its revenue (qty × price), highest revenue first (two columns).	<pre>SELECT st.city, SUM(s.qty * p.price) AS revenue FROM sales s JOIN stores st ON s.store_id = st.store_id JOIN products p ON s.sku = p.sku GROUP BY st.city ORDER BY revenue DESC;</pre> Any SELECT returning the same rows in the same order (column order does not matter) is accepted.
8	WHERE hands-on	List the sku of every product whose in_stock is below its reorder_at.	<pre>SELECT sku FROM products WHERE in_stock < reorder_at;</pre> Any SELECT returning the same rows (column order does not matter) is accepted.
9	Integer division concept	In SQLite, what is SELECT 7 / 2;?	3 Other options: 3.5 · 4 · An error
10	NULL concept	Which WHERE clause finds rows where qty is missing?	WHERE qty IS NULL Other options: WHERE qty = NULL · WHERE qty = '' · WHERE qty = 0
11	UPDATE safely concept	What does UPDATE products SET price = 0; do?	Sets the price of EVERY product to 0 Other options: Nothing — WHERE is required · Sets only the first product · Shows an error
12	HAVING concept	What is the difference between WHERE and HAVING?	WHERE filters rows before grouping; HAVING filters groups after Other options: They are the same · HAVING only works with text · WHERE only works with JOIN
13	Aggregates concept	The products table has one row per product and there are 24 products. What does SELECT COUNT(*) FROM products; return?	24
14	JOIN concept	Which spreadsheet idea is most like a SQL JOIN?	A VLOOKUP/XLOOKUP for every row Other options: SUM · Sorting · Conditional formatting
15	Syntax concept	How is a text value written in standard SQL?	'Snacks' (single quotes) Other options: "Snacks" (double quotes) · Snacks (no quotes) · [Snacks]

Discussion

1. Which spreadsheet function is most like GROUP BY? Like JOIN? Like WHERE?

2. Why do real databases not have an Undo button — and what do companies do instead?

Exit ticket

Write SQL for the number of sales per store.

Answer: `SELECT store_id, COUNT(*) FROM sales GROUP BY store_id;`

Lesson 8: Cross-Platform Translator

Workspace: **Compare** · 5 challenges · about 25 minutes

The same formula can work, fail, or silently give a different answer depending on the app. Checking before you share is a professional habit.

Students will be able to

- Which functions exist where
- Silent differences between apps
- Spreadsheet ideas in SQL

Common misconceptions

- Test your file in the app your audience will use.
- Prefer functions that exist everywhere when sharing widely.

Vocabulary

Compatibility	Whether a formula works in another app or version.	<code>XLOOKUP: Excel 365 ✓, Excel 2013 ✗</code>
Silent difference	No error, but a different result — the most dangerous kind.	<code>SORT(..., -1)</code> in Sheets
Translation	Rewriting a formula so it works elsewhere.	<code>XLOOKUP → INDEX/MATCH</code>

Challenges & answer key

#	Challenge	Task	Answer
1	Who Says #NAME? ★☆☆ · 10 XP	Where does <code>=XLOOKUP(A2, B:B, C:C)</code> give a <code>#NAME?</code> error? <i>Takeaway: Excel 2013 never got XLOOKUP. INDEX/MATCH works everywhere.</i>	Answer: Excel 2013
2	The Silent Sort ★★★ · 15 XP	You type <code>=SORT(A2:C20, 3, -1)</code> in Google Sheets. What happens? <i>Takeaway: No error, wrong answer. Always check your results!</i>	Answer: It sorts low → high
3	Pointing at Nothing ★☆☆ · 10 XP	Cell A1 is empty. What does <code>=A1</code> show in Excel vs Google Sheets? <i>Takeaway: Little differences like this break checks such as <code>=IF(B2="", ...)</code>.</i>	Answer: Excel: 0, Sheets: blank

4	Array Formulas Through Time ★☆☆ · 15 XP	Which one needs Ctrl+Shift+Enter (or ARRAYFORMULA) to make =MAX(IF(C2:C25="Snacks", D2:D25)) give the right answer? Takeaway: Excel 2013 needs Ctrl+Shift+Enter; Google Sheets needs ARRAYFORMULA (Ctrl+Shift+Enter adds it for you).	Answer: Excel 2013 and Google Sheets
5	Spreadsheet → SQL ★☆☆ · 10 XP	Which SQL clause does the same job as VLOOKUP? Takeaway: JOIN matches rows from two tables, like a lookup on every row at once.	Answer: JOIN

Certification test bank (13 questions; each test draws 10, half hands-on; 80% to pass)

#	Topic	Question	Answer
1	Translation hands-on	Translate =XLOOKUP("SKU-502",Products!A2:A25,Products!B2:B25) so it works in Excel 2013.	<code>=INDEX(Products!B2:B25, MATCH("SKU-502",Products!A2:A25,0))</code> Excel 2013 · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
2	Silent differences hands-on	Excel 365's =SORT(Products!B2:D25,3,-1) sorts by price, highest first. Write the version that does the same in Google Sheets.	<code>=SORT(Products!B2:D25,3,FALSE)</code> Google Sheets · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
3	Translation hands-on	Translate =MAXIFS(Products!D2:D25,Products!C2:C25,"Electronics") for Excel 2013 (highest Electronics price).	<code>=MAX(IF(Products!C2:C25="Electronics",Products!D2:D25))</code> <i>Ctrl+Shift+Enter</i> Excel 2013 · Scratch!B2. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
4	Translation hands-on	Excel 365's =CONCAT(B2," ",C2) breaks in Google Sheets. In Stores!G2, write a Sheets version (like Cumming, GA). Filled down to row 6.	<code>=B2&" ", "&C2</code> Google Sheets · Stores!G2 filled to row 6. Any formula with the same results on the original and on a shuffled copy of the data is accepted.

5	Translation hands-on	Translate =IFS(D2>=10,"High",D2>=3,"Mid",TRUE,"Low") for Excel 2013, in Products!J2. Filled down to row 25.	<pre>=IF(D2>=10, "High", IF(D2>=3, "Mid", "Low"))</pre> Excel 2013 · Products!J2 filled to row 25. Any formula with the same results on the original and on a shuffled copy of the data is accepted.
6	Spreadsheet → SQL hands-on	Write the SQL version of =SUMIF(Sales!C2:C61,"S01",Sales!E2:E61) (one value).	<pre>SELECT SUM(qty) FROM sales WHERE store_id = 'S01' ;</pre> Any SELECT returning the same rows (column order does not matter) is accepted.
7	Compatibility concept	Which function works the same in Excel 365, Excel 2013 AND Google Sheets?	SUMIF Other options: XLOOKUP · TEXTJOIN · LET
8	Silent differences concept	Why is a "silent difference" between apps more dangerous than an error?	You get a wrong answer and nothing warns you Other options: It crashes the app · It deletes the file · It only happens on phones
9	Errors concept	You type a broken formula. What do Excel and Google Sheets do?	Excel refuses it with a pop-up; Sheets accepts it and shows #ERROR! Other options: Both show #ERROR! · Both show a pop-up · Excel shows #ERROR!; Sheets shows a pop-up
10	Spreadsheet → SQL concept	SQL's GROUP BY store_id with SUM(qty) is most like...	A SUMIF for every store at once Other options: A VLOOKUP · Sorting by store · Deleting duplicates
11	Compatibility concept	Where does the spill reference B2# work?	Only in Excel 365 Other options: Excel 365 and Excel 2013 · Google Sheets only · All three
12	Compatibility concept	You must share a lookup with people who might use Excel 2013 or Google Sheets. Which is safest?	INDEX + MATCH Other options: XLOOKUP · FILTER · LET
13	Compatibility concept	What does Excel 2013 show for =UNIQUE(C2:C25)?	#NAME? or #name

Discussion

1. Before sharing a spreadsheet, what should you check?

Exit ticket

Name a function that exists in Excel 365 but gives #NAME? in Excel 2013.

Answer: `XLLOOKUP`, `FILTER`, `SORT`, `UNIQUE`, `IFS`, `TEXTJOIN`, `LET`, `MAXIFS...`

Lesson 9: Databases 101: What is an RDBMS?

Workspace: **RDBMS Explorer** · 10 challenges · about 60 minutes

Spreadsheets hold data; a relational database guards it. Real companies — and the AI systems they build — keep their important data in an RDBMS because it refuses bad data, connects tables with keys, and never leaves a change half-done.

Students will be able to

- Tables, rows, columns and data types
- Primary keys and foreign keys
- Relationships and JOINS
- Constraints that refuse bad data
- Transactions and ACID

Common misconceptions

- A spreadsheet that repeats the same fact in many rows will eventually disagree with itself.
- A foreign key must point at a row that already exists — add the parent first.
- Nothing inside BEGIN is permanent until COMMIT.

Vocabulary

Table / row / column	A table holds one kind of thing; each row is one item; each column is one typed attribute.	<code>products: one row per product</code>
Primary key	A column whose value is different for every row — the row's ID.	<code>products.sku</code>
Foreign key	A column that holds another table's primary key, linking the two.	<code>sales.sku → products.sku</code>
Constraint	A rule the database enforces: NOT NULL, UNIQUE, CHECK, FOREIGN KEY, types.	<code>CHECK (price >= 0)</code>
Normalization	Store each fact once, in one place, and link to it.	<code>Price lives only in products</code>
Transaction	Several changes that succeed or fail together.	<code>BEGIN; ... COMMIT; (or ROLLBACK;)</code>

Challenges & answer key

#	Challenge	Task	Answer
1	The R in RDBMS ☆☆☆ · 10 XP	What does the R in RDBMS stand for? Takeaway: Relational: data lives in tables, and keys create RELATIONSHIPS between	Answer: Relational

them.

2	Break the Spreadsheet ☆☆☆ · 15 XP	In section 1. Grid vs. database, change the price in only ONE of the highlighted rows of the flat sheet. Watch the sheet start to disagree with itself. Takeaway: When the same fact is stored many times, copies drift apart. That is an "update anomaly". Databases avoid it by storing each fact once.	Section 1: change only one of the highlighted Price cells in the flat sheet.
3	Pick a Primary Key ☆☆☆ · 10 XP	You are building a students table for your school. Which column makes the best primary key? Takeaway: Names, grades and birthdays repeat. A student ID is unique — perfect for a primary key.	Answer: Student ID number
4	Follow the Key ☆☆☆ · 10 XP	sales.store_id holds values like S03 that must already exist in stores.store_id. What is sales.store_id? Takeaway: A foreign key points at another table's primary key. Following it is what a JOIN does.	Answer: A foreign key
5	The Database Says NO ☆☆☆ · 20 XP	Make the database refuse bad data in 3 different ways (duplicate key, missing value, broken rule, missing parent, wrong type...). Use section 4 or any console on this page. Takeaway: Constraints stop bad data at the door. In a spreadsheet you only find these problems later — while cleaning.	<pre>INSERT INTO products VALUES ('SKU-101', 'Fake', 'Snacks', 1, 0.4, 10, 5, 'X');</pre> <pre>INSERT INTO products (sku, category, price, in_stock) VALUES ('SKU-999', 'Snacks', 1, 5);</pre> <pre>UPDATE products SET price = -5 WHERE sku = 'SKU-101';</pre> Any 3 of the 6 "Try it" buttons in section 4 (UNIQUE, NOT NULL, CHECK, FOREIGN KEY, type).
6	Build Your Own Table ☆☆☆ · 25 XP	In Free play, create a table students with student_id INTEGER PRIMARY KEY and name TEXT NOT NULL, then insert at least 2 students. Takeaway: An INTEGER PRIMARY KEY numbers new rows automatically — no duplicates, ever.	<pre>CREATE TABLE students (student_id INTEGER PRIMARY KEY, name TEXT NOT NULL);</pre> <pre>INSERT INTO students (name) VALUES ('Ana'), ('Ben');</pre>
7	Connect Two Tables ☆☆☆ · 30 XP	Create enrollments with student_id INTEGER REFERENCES students(student_id) and course TEXT NOT NULL. Add an enrollment for one	<pre>CREATE TABLE students (student_id INTEGER PRIMARY KEY, name TEXT NOT NULL);</pre>

of your students. (Then try enrolling student 99 and watch it fail!)

Takeaway: You just built a relationship: one student, many enrollments — and the database guarantees every enrollment belongs to a real student.

```
INSERT INTO students (name)
VALUES ('Ana'), ('Ben');
```

```
CREATE TABLE enrollments
(student_id INTEGER REFERENCES
students(student_id), course
TEXT NOT NULL);
```

```
INSERT INTO enrollments VALUES
(1, 'Intro to AI');
```

Build students first (previous challenge).

8 **Undo With ROLLBACK**
★★★ · 20 XP

In section 5, start a transaction, DELETE every sale, check they are gone — then ROLLBACK and bring all 60 back.

Takeaway: Inside a transaction nothing is permanent until COMMIT. ROLLBACK is a real undo button for the whole database.

```
BEGIN; DELETE FROM sales;
SELECT COUNT(*) FROM sales;
ROLLBACK;
```

9 **Survive the Power Failure**
★★★ · 25 XP

In the lunch-money demo (section 5), start with a transaction, pull the plug with ⚡ Power failure!, then do the whole transfer again and COMMIT it.

Takeaway: Atomicity: a transfer is either completely done or not done at all — money never vanishes.

Section 5: With a transaction (BEGIN) → Step 1 → ⚡ Power failure! → With a transaction (BEGIN) → Step 1 → Step 2 → COMMIT.

10 **All or Nothing**
★★★ · 10 XP

In a database transaction, atomic means...

Takeaway: Atomic = indivisible. ACID: Atomic, Consistent, Isolated, Durable.

Answer: Either every change happens or none of them do

Certification test bank (16 questions; each test draws 10, half hands-on; 80% to pass)

#	Topic	Question	Answer
1	Primary keys / NOT NULL hands-on	Create a table clubs with club_id (a whole-number primary key the database fills in automatically) and name (text that can never be missing).	<pre>CREATE TABLE clubs (club_id INTEGER PRIMARY KEY, name TEXT NOT NULL);</pre> Graded by behavior: hidden test statements must succeed or be refused exactly as with this answer.
2	Foreign keys hands-on	Create two tables: clubs (club_id INTEGER primary key, name TEXT) and members (member_id INTEGER primary key, name TEXT, club_id) — where a member's club_id must point to a club that exists.	<pre>CREATE TABLE clubs (club_id INTEGER PRIMARY KEY, name TEXT); CREATE TABLE members (member_id INTEGER PRIMARY KEY, name TEXT, club_id INTEGER REFERENCES clubs(club_id));</pre> Graded by behavior: hidden test statements must succeed or be refused exactly as with this answer.

3	CHECK constraints hands-on	Create a table grades with student (text, never missing) and score (a whole number that must be between 0 and 100, inclusive).	<pre>CREATE TABLE grades (student TEXT NOT NULL, score INTEGER CHECK (score >= 0 AND score <= 100));</pre> Graded by behavior: hidden test statements must succeed or be refused exactly as with this answer.
4	UNIQUE hands-on	Create a table accounts with id (INTEGER primary key) and email — text that is required and can never be used twice.	<pre>CREATE TABLE accounts (id INTEGER PRIMARY KEY, email TEXT NOT NULL UNIQUE);</pre> Graded by behavior: hidden test statements must succeed or be refused exactly as with this answer.
5	DEFAULT hands-on	Create a table tickets: ticket_id (INTEGER primary key), title (required text) and status (text that is 'open' when nobody gives one).	<pre>CREATE TABLE tickets (ticket_id INTEGER PRIMARY KEY, title TEXT NOT NULL, status TEXT DEFAULT 'open');</pre> Graded by behavior: hidden test statements must succeed or be refused exactly as with this answer.
6	Transactions hands-on	Raise the price of every Snacks product by 10%, as one transaction (BEGIN ... COMMIT) so the change is saved for good.	<pre>BEGIN; UPDATE products SET price = price * 1.1 WHERE category = 'Snacks'; COMMIT;</pre> Graded by behavior: hidden test statements must succeed or be refused exactly as with this answer. Must use BEGIN and COMMIT.
7	Keys & JOIN hands-on	Use the foreign key sales.store_id → stores.store_id to list every order_id with its store's manager (two columns).	<pre>SELECT s.order_id, st.manager FROM sales s JOIN stores st ON s.store_id = st.store_id;</pre> Any SELECT returning the same rows (column order does not matter) is accepted.
8	Primary keys concept	What is a primary key?	A column whose value is different for every row, used as the row's ID Other options: The first column of every table · A password for the database · A column that links to another table
9	Foreign keys concept	sales.sku REFERENCES products.sku. What does the database do with a sale for SKU-999 (no such product)?	Refuses the INSERT Other options: Creates SKU-999 in products · Saves it with a warning · Changes it to NULL
10	Normalization concept	A flat sheet repeats each product's price on every order row. Someone	An update anomaly

		fixes the price on only one row. This is called...	Other options: A foreign key · A transaction · Normalization
11	Transactions concept	Halfway through a money transfer inside BEGIN ... COMMIT, the power fails. What happens?	The unfinished transaction is undone — no money is lost Other options: Half the transfer is saved · The database is destroyed · The money is doubled
12	What is an RDBMS concept	Which of these is NOT a relational database management system?	A CSV file Other options: PostgreSQL · MySQL · SQLite
13	Data types concept	products.price is a REAL column in a STRICT table. What happens with UPDATE products SET price = 'cheap'?	The database refuses it Other options: The price becomes 0 · It is stored as text · Every price becomes "cheap"
14	Normalization concept	What does normalization mean?	Store each fact once, in one place, and link to it with keys Other options: Make all numbers between 0 and 1 · Sort every table · Delete NULL values
15	Transactions concept	After BEGIN; DELETE FROM sales; you run ROLLBACK;. How many sales rows are left (there were 60)?	60 Other options: 0 · 59 · It depends on the computer
16	Transactions concept	In ACID, what does the A (atomicity) promise?	All of a transaction's changes happen, or none do Other options: Data is encrypted · Queries run fast · Tables have primary keys

Discussion

1. Your school keeps student records. Why would it use a database instead of one giant spreadsheet?
2. Give an example from your own life where "all or nothing" matters (like the lunch-money transfer).
3. Constraints stop bad data before it is saved. How would that have helped in the Data Wrangling Lab?

Exit ticket

Write the CREATE TABLE for clubs(club_id, name) where club_id is the primary key and name is required.

Answer: `CREATE TABLE clubs (club_id INTEGER PRIMARY KEY, name TEXT NOT NULL);`

Lesson 10: Databases for Machine Learning

Workspace: **ML Data Lab** · 12 challenges · about 75 minutes

A machine-learning model only knows what its training table shows it. Most AI mistakes are really data mistakes: the wrong rows, a column that gives away the answer, or a test that was secretly part of the training. SQL is how data scientists find them.

Students will be able to

- Examples, features and labels
- Class balance and baselines
- Missing values
- Train/test splits and data leakage
- Grading a model: accuracy and the confusion matrix

Common misconceptions

- High accuracy can be meaningless when one answer is rare — compare with a baseline.
- If a model looks perfect, look for leakage before celebrating.
- Never let test rows (or copies of them) into the training table.

Vocabulary

Example	One row the model learns from.	Product SKU-101 at store S03 in week 2
Feature	An input column the model may use to make its guess.	promo, price, units_last_week
Label	The answer the model is trying to predict.	stockout (1 = ran out, 0 = did not)
Class imbalance	One answer is much rarer than the other.	17 stockouts vs 47 non-stockouts
Train / test split	Learn from some rows; check on rows the model has never seen.	WHERE split = 'test'
Data leakage	The model sees the answer during training — test copies, or a column only known afterwards.	restock_after
Baseline	The score of a "dumb" model. A real model must beat it.	Always predict 0
Confusion matrix	Counts of right and wrong guesses for each answer.	GROUP BY stockout, predicted

Challenges & answer key

#	Challenge	Task	Answer
1	Find the Label ☆☆☆ · 10 XP	The model will predict whether a product runs out this week. Which column of ml_examples is the label? Takeaway: The label is the answer column. Everything the model may look at to guess it is a feature.	Answer: stockout
2	Sort the Columns ☆☆☆ · 20 XP	In section 1, give every column its job — ID, feature, label, split, or "known only afterwards" — and press Check my sorting. Takeaway: Before any AI work, decide what each column is for. IDs and after-the-fact columns must never be features.	Section 1: example_id → Row ID; week → Train/test bookkeeping (Feature also accepted); store_id, sku, units_last_week, promo, price, in_stock_start → Feature; restock_after → Known only afterwards; stockout → Label; split → Train/test bookkeeping. Then "Check my sorting".
3	Count the Classes ☆☆☆ · 15 XP	Write a query that shows how many examples have each stockout value (0 and 1). Takeaway: Only about 1 in 4 examples is a stockout. With imbalanced classes, "accuracy" alone can fool you.	<pre>SELECT stockout, COUNT(*) FROM ml_examples GROUP BY stockout;</pre>
4	Hunt the Missing Values ☆☆☆ · 15 XP	Write a query that counts the examples where units_last_week is missing. Takeaway: Models cannot use NULL directly. You must drop those rows, fill them in (for example with the average), or add a "was missing" flag — and write down which you chose.	<pre>SELECT COUNT(*) FROM ml_examples WHERE units_last_week IS NULL;</pre>
5	Check the Split ☆☆☆ · 15 XP	Show how many examples are in each split (train and test). Takeaway: The test rows are the model's final exam. It must never study them.	<pre>SELECT split, COUNT(*) FROM ml_examples GROUP BY split;</pre>
6	Catch the Leak ☆☆☆ · 30 XP	Some test rows are copies of training rows (same week, store_id and sku). Write a query that lists the week, store_id and sku of every example that appears in both splits. Takeaway: If the model is tested on rows it already studied, its test score is a lie. Removing duplicates across splits is a basic step in every real ML project.	<pre>SELECT week, store_id, sku FROM ml_examples GROUP BY week, store_id, sku HAVING COUNT(DISTINCT split) > 1;</pre>
7	Too Good to Be True ☆☆☆ · 15 XP	Which column would let a model "cheat", because it is only known after	Answer: restock_after

the week is over?

Takeaway: restock_after is filled in after a stockout happens. Using it is target leakage: perfect in testing, useless in real life.

8 **Watch a Model Cheat**
★☆☆ · 15 XP

In section 4, run the model that uses restock_after and see its "perfect" score.

Takeaway: A perfect score is a warning sign. Real models are never perfect — check for leakage first.

Section 4: press "Train a model that uses restock_after".

9 **Build the Training Table**
★★★ · 35 XP

Create a table train_set with only the training rows, without the leaky restock_after column, and with the product's category joined in from products.

Takeaway: A "feature table" is a JOIN that gathers every input the model may use — and nothing it must not see.

```
CREATE TABLE train_set AS
SELECT e.example_id, e.promo,
e.price, e.units_last_week,
e.in_stock_start, p.category,
e.stockout FROM ml_examples e
JOIN products p ON e.sku = p.sku
WHERE e.split = 'train';
```

10 **Grade the Model**
★★★ · 25 XP

The predictions table holds a model's guesses for the test rows. Write ONE query that returns the model's accuracy: the fraction (or percent) of predictions that match the real stockout.

Takeaway: Accuracy = right guesses ÷ all guesses. It is the first number everyone asks about — and often not the most important one.

```
SELECT AVG(p.predicted =
e.stockout) FROM predictions p
JOIN ml_examples e ON
p.example_id = e.example_id;
Also accepted: 100.0 * SUM(p.predicted
= e.stockout) / COUNT(*) (a percent).
```

11 **Build a Confusion Matrix**
★★★ · 30 XP

Write a query that counts the test predictions for every combination of real stockout and predicted value (columns: stockout, predicted, count).

Takeaway: The confusion matrix shows WHICH mistakes a model makes: missed stockouts (false negatives) and false alarms (false positives).

```
SELECT e.stockout, p.predicted,
COUNT(*) FROM predictions p JOIN
ml_examples e ON p.example_id =
e.example_id GROUP BY
e.stockout, p.predicted;
```

12 **Beat the Baseline**
★★★ · 15 XP

On the test rows, "always predict 0 (no stockout)" is right 83% of the time. The real model is right 75% of the time. What does this tell you?

Takeaway: Always compare with a baseline, and look at the confusion matrix. A model that never predicts the rare class can still have high accuracy.

Answer: With rare stockouts, accuracy is misleading: the model is worse than doing nothing

#	Topic	Question	Answer
1	Class balance hands-on	Show how many examples have each stockout value (two columns: stockout, count).	<pre>SELECT stockout, COUNT(*) FROM ml_examples GROUP BY stockout;</pre> Any SELECT returning the same rows (column order does not matter) is accepted.
2	Missing values hands-on	Count the examples whose units_last_week is missing (one value).	<pre>SELECT COUNT(*) FROM ml_examples WHERE units_last_week IS NULL;</pre> Any SELECT returning the same rows (column order does not matter) is accepted.
3	Train/test split hands-on	Show how many examples are in each split (two columns: split, count).	<pre>SELECT split, COUNT(*) FROM ml_examples GROUP BY split;</pre> Any SELECT returning the same rows (column order does not matter) is accepted.
4	Data leakage hands-on	List the week, store_id and sku of every combination that appears in both the train and the test split (three columns).	<pre>SELECT week, store_id, sku FROM ml_examples GROUP BY week, store_id, sku HAVING COUNT(DISTINCT split) > 1;</pre> Any SELECT returning the same rows (column order does not matter) is accepted.
5	Accuracy hands-on	Return the model's accuracy on the test rows as a fraction between 0 and 1 (one value). The model's guesses are in predictions.	<pre>SELECT AVG(p.predicted = e.stockout) FROM predictions p JOIN ml_examples e ON p.example_id = e.example_id;</pre> Any SELECT returning the same rows (column order does not matter) is accepted.
6	Confusion matrix hands-on	Build the confusion matrix: one row per (real stockout, predicted) pair that occurs, with its count (three columns).	<pre>SELECT e.stockout, p.predicted, COUNT(*) FROM predictions p JOIN ml_examples e ON p.example_id = e.example_id GROUP BY e.stockout, p.predicted;</pre> Any SELECT returning the same rows (column order does not matter) is accepted.
7	Missing values hands-on	Return the average units_last_week of the training rows, skipping missing values (one value).	<pre>SELECT AVG(units_last_week) FROM ml_examples WHERE split = 'train';</pre> Any SELECT returning the same rows (column order does not matter) is

			accepted.
8	Missing values hands-on	List every example_id with its units_last_week, using 0 where the value is missing (two columns).	<pre>SELECT example_id, COALESCE(units_last_week, 0) FROM ml_examples;</pre> Any SELECT returning the same rows (column order does not matter) is accepted.
9	Class balance hands-on	Return the fraction of training examples that are stockouts (a number between 0 and 1).	<pre>SELECT AVG(stockout) FROM ml_examples WHERE split = 'train';</pre> Any SELECT returning the same rows (column order does not matter) is accepted.
10	Feature tables hands-on	Create a table train_set holding only the training rows, without restock_after, keeping the label stockout, and with each product's category joined in from products.	<pre>CREATE TABLE train_set AS SELECT e.example_id, e.promo, e.price, e.units_last_week, e.in_stock_start, p.category, e.stockout FROM ml_examples e JOIN products p ON e.sku = p.sku WHERE e.split = 'train';</pre> Graded by behavior: hidden test statements must succeed or be refused exactly as with this answer.
11	Features and labels concept	In a machine-learning table, what is the label?	The answer the model is trying to predict Other options: The name of the table · Any text column · The primary key
12	Features and labels concept	Why should example_id NOT be used as a feature?	It is just a row number and says nothing about the real world Other options: It is too large · It is a foreign key · Integers cannot be features
13	Data leakage concept	A model that uses restock_after scores 100% in testing but fails in real life. Why?	restock_after is only known after the stockout happens — target leakage Other options: The test set was too small · The model needed more features · restock_after has missing values
14	Data leakage concept	Some test rows are copies of training rows. What happens to the test score?	It looks better than the model really is Other options: It looks worse than the model really is · Nothing changes · The database refuses to run
15	Baselines concept	Only 1 in 6 test rows is a stockout. A model that ALWAYS predicts "no	No — it never catches a single stockout

		stockout" gets about 83% accuracy. Is it useful?	Other options: Yes — 83% is a B · Yes — it is never wrong about non-stockouts, so it is perfect · It depends on the computer
16	Confusion matrix concept	The real stockout was 1, but the model predicted 0. What kind of mistake is that?	A false negative (a missed stockout) Other options: A false positive (a false alarm) · A true negative · A true positive
17	Train/test split concept	Why are weeks 1–4 used for training and weeks 5–6 for testing, instead of mixing all weeks?	A real model predicts the future, so it should be tested on later weeks it never saw Other options: Later weeks have more rows · SQL can only sort by week · It makes the label balanced
18	Accuracy concept	A model is right on 18 of 24 test rows. What is its accuracy, in percent? (number only)	75 or 75%

Discussion

1. A model scores 99% accuracy at spotting a rare disease that 1% of people have. Should you trust it? What would you ask to see?
2. Where could leakage sneak into a school project — for example, predicting final grades?
3. Missing values: when is it OK to drop the rows, and when does that make the model unfair?

Exit ticket

Write SQL that counts how many TEST rows are stockouts.

Answer: `SELECT COUNT(*) FROM ml_examples WHERE split = 'test' AND stockout = 1;`

Appendix: what is different in each tool

Excel 365

- Dynamic arrays: one formula can **spill** results into the cells below it. If something is in the way you get `#SPILL!`.
- Has XLOOKUP, XMATCH, FILTER, SORT, UNIQUE, SEQUENCE, LET, IFS, SWITCH, TEXTJOIN, CONCAT, MAXIFS and MINIFS.
- Refer to a whole spilled result with `A2#` (the spill operator). Google Sheets does not have this.
- `SORT(range, 3, -1)` sorts column 3 high→low. Sort order is **1 or -1**.
- `FILTER(range, include, [if_empty])` takes ONE include test. Combine tests with `*` (AND) or `+` (OR).
- Open-ended ranges like `A2:A` are **not** allowed. Use `A:A` or `A2:A1000`.
- A formula pointing at an empty cell shows **0**.
- Excel refuses to accept a formula with a typo and shows a **"There's a problem with this formula"** pop-up.

Excel 2013

- **No** XLOOKUP, FILTER, SORT, UNIQUE, LET, IFS, SWITCH, TEXTJOIN, CONCAT, MAXIFS, MINIFS → typing them gives `#NAME?`.
- Lookups: use `VLOOKUP(..., FALSE)` or the classic combo `INDEX(return_range, MATCH(value, lookup_range, 0))`.
- Instead of IFS, nest IFs: `IF(test1, a, IF(test2, b, c))`.
- No spilling. A formula that works on a whole range must be confirmed with **Ctrl+Shift+Enter**. Excel shows it as `{=MAX(IF(...))}`.
- Without Ctrl+Shift+Enter, a range used where one value is expected is cut down to the value in the **same row** (implicit intersection) — often the wrong answer!
- `SUMPRODUCT` handles arrays without Ctrl+Shift+Enter, which is why it was so popular.
- Join text with `CONCATENATE(a, b, c)` or `a & b & c`.

Google Sheets

- `FILTER(range, condition1, condition2, ...)` — each condition is its **own argument**, and there is **no if_empty**. No matches → `#N/A`.
- `SORT(range, column, is_ascending)` uses **TRUE/FALSE**. Typing Excel's `-1` still sorts ascending (any non-zero number counts as TRUE)!
- Operators on whole ranges (like `D2:D25*F2:F25`) need `ARRAYFORMULA( )`. Ctrl+Shift+Enter adds it for you.
- `CONCAT` only joins **two** values. Use `CONCATENATE`, `JOIN`, `TEXTJOIN` or `&` for more.
- Open-ended ranges like `A2:A` work (from row 2 to the bottom).
- Sheets-only functions: QUERY, ARRAYFORMULA, SPLIT, JOIN, REGEXMATCH, REGEXEXTRACT, REGEXREPLACE, COUNTUNIQUE, IMPORTRANGE, GOOGLEFINANCE.
- A bad formula is still saved — the cell shows `#ERROR!` (parse error) instead of a pop-up.
- A formula pointing at an empty cell shows a **blank**, not 0.

CSV / TSV

- A CSV/TSV file stores **values only**. Saving as CSV turns every formula into its current result.
- No formatting (bold, colors, \$ signs as formats), and only **one sheet** per file.
- A value that contains a comma must be wrapped in "double quotes": "Gel Pens, 4-pack".
- A double quote inside a quoted value is written twice: "12"" Ruler".
- TSV uses the TAB character, so commas inside values are no problem.
- Spreadsheets "helpfully" convert values when they open a CSV: 05401 → 5401, long numbers → 4.41712E+15.

SQL Database

- Text goes in '**single quotes**'. "Double quotes" mean a column name.
- Text comparisons are **case-sensitive**: 'snacks' does not equal 'Snacks'. LIKE is not case-sensitive.
- Dividing two whole numbers drops the decimals: 7 / 2 = 3. Use 7 / 2.0 or AVG().
- Columns are referred to by name, not letter: price, not D.
- Joining tables (JOIN) replaces VLOOKUP. Grouping (GROUP BY) replaces SUMIF/COUNTIF.
- UPDATE and DELETE without WHERE change **every row**. There is no undo in a real database!

Appendix: error codes

#NAME?	The formula contains unrecognized text (usually a misspelled or unsupported function name).
#VALUE!	Wrong type of value — e.g. doing math on text, or a range where one value was expected.
#REF!	The formula refers to a cell or sheet that does not exist.
#DIV/0!	Dividing by zero (or by an empty cell).
#N/A	A value is not available — usually a lookup that found no match.
#NUM!	A number is invalid or too big/small.
#NULL!	Ranges do not intersect.
#SPILL!	The result is an array, but cells it needs to spill into are not empty.
#CALC!	The calculation engine hit something it does not support, like an empty array.
#ERROR!	Google Sheets could not understand (parse) the formula.